# SUPPLEMENTARY MATERIALS


**Authors:**

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**Title:**

Lentivirus or AAV viral vector mediated gene therapy interventions in ischemic stroke: a systematic review of preclinical *in vivo* studies


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**Journal name:** Journal of Cerebral Blood Flow & Metabolism


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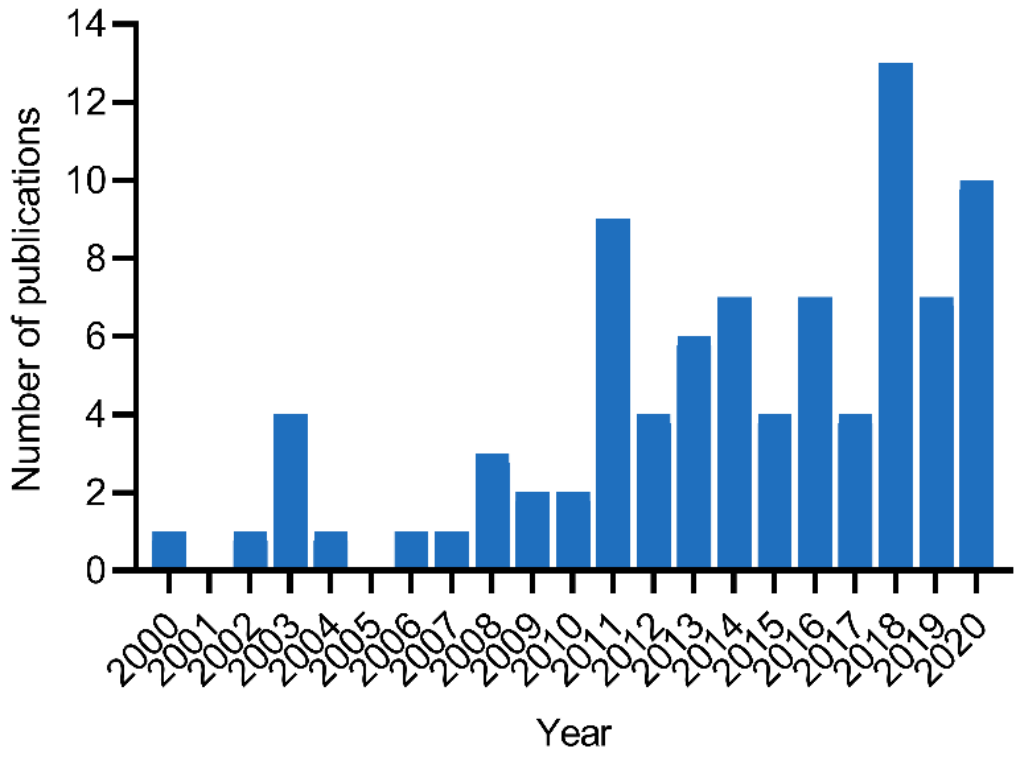
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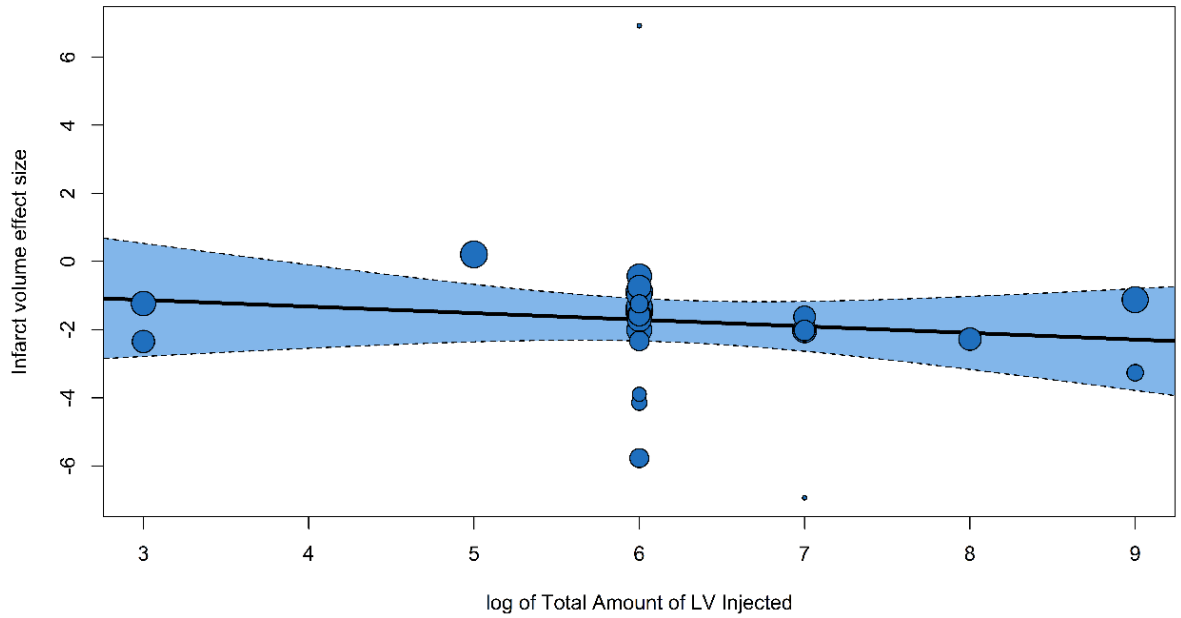
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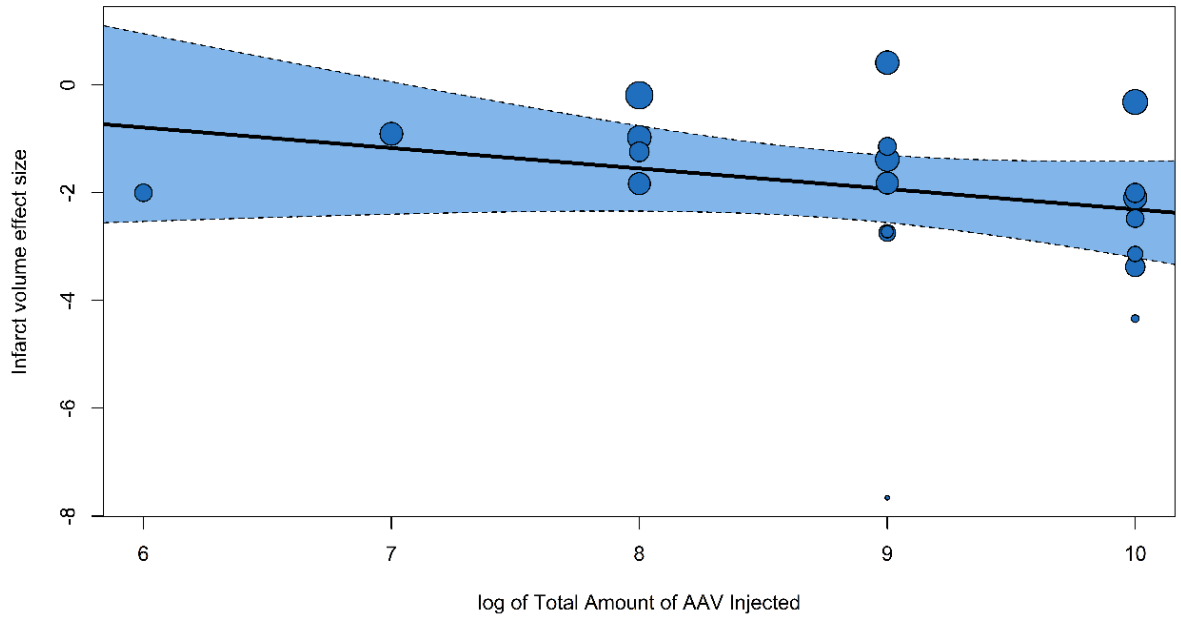
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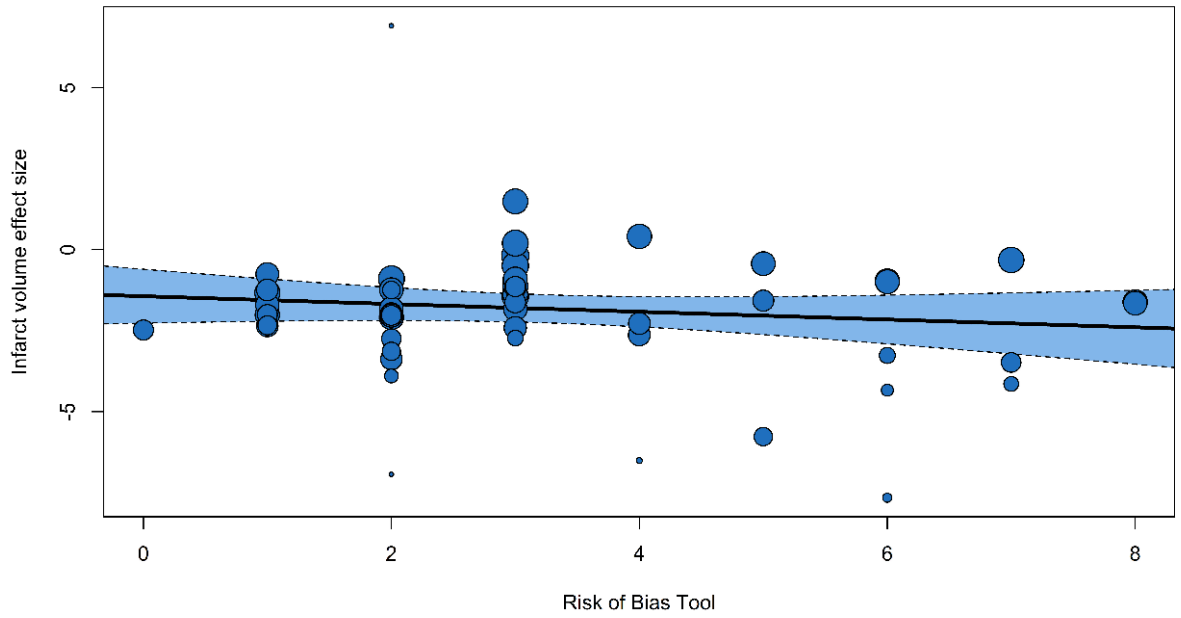

**Supplementary Figure 1.** Number of included publications per year



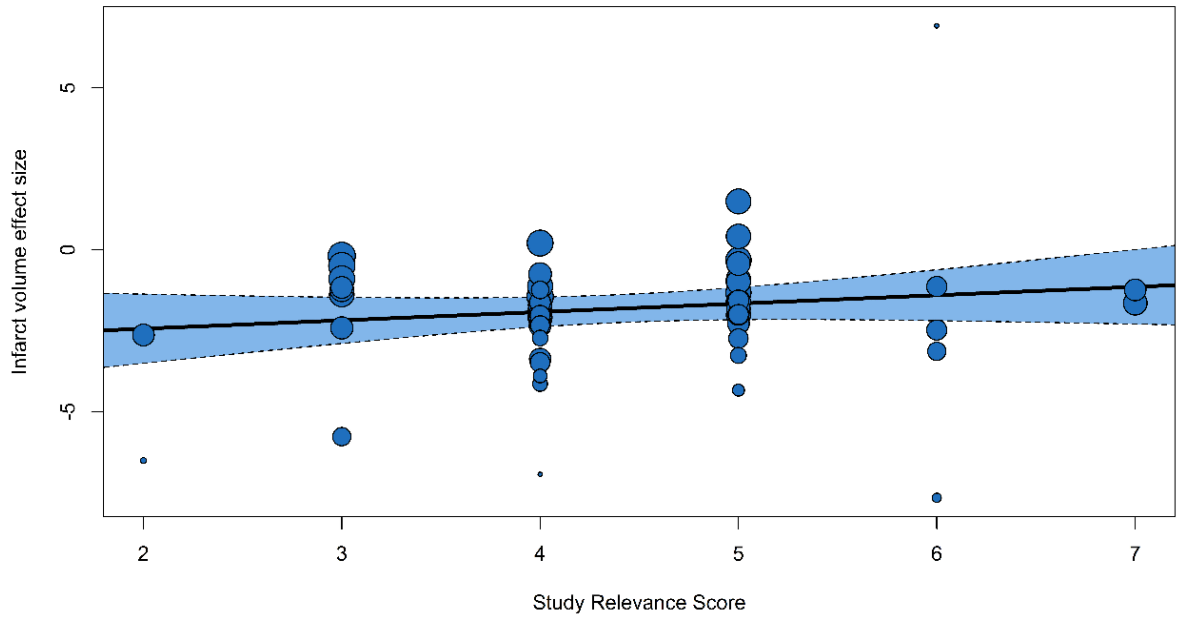

**Supplementary Figure 2. Metaregression analysis of lentivirus vector dose and infarct volume effect size.** Scatterplot shows no correlation between effect size and LV dose ($P = 0.4322$. The units on the *x*-axis represent exponent symbols with the base 10 (e.g. $10^3$)). Shaded area corresponds to 95% confidence interval.



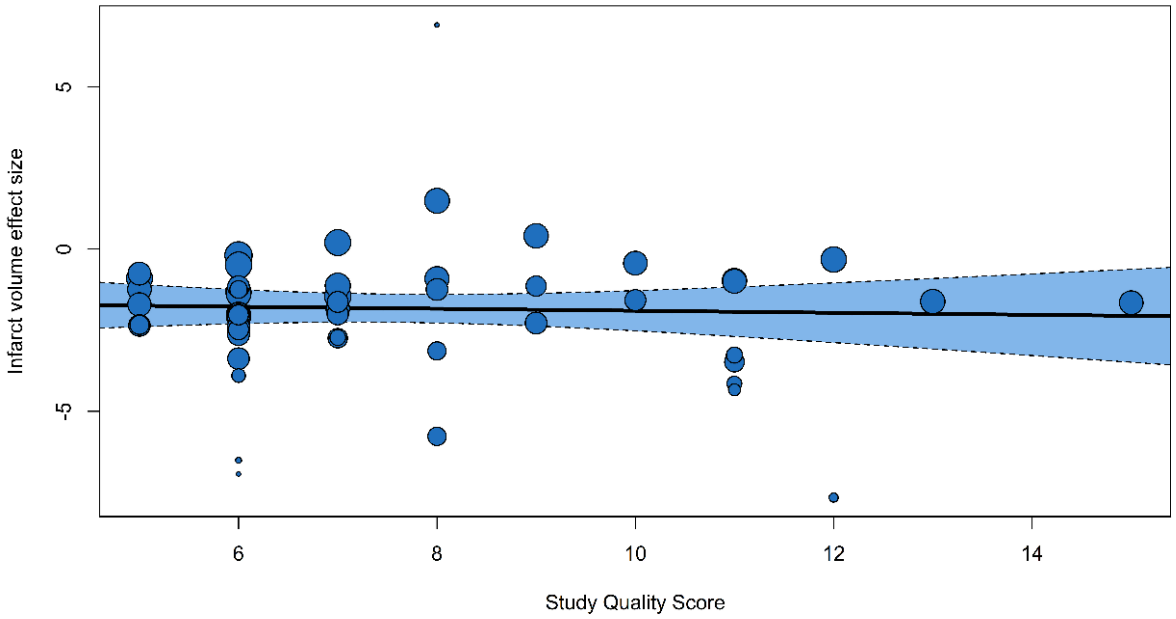

**Supplementary Figure 3. Metaregression analysis of adeno-associated virus vector dose and infarct volume effect size.** Scatterplot shows no correlation between effect size and AAV dose (*P* = 0.1903). The units on the *x*-axis represent exponent symbols with the base 10 (e.g. $10^6$). Shaded area corresponds to 95% confidence interval.




**Supplementary Figure 4. Metaregression analysis of risk of bias score and infarct volume effect size.** Scatterplot shows no correlation between effect size and risk of bias score (*P* = 0.2874). The values on the *x*-axis represents risk of bias score. Higher score represent lower risk of bias. Shaded area corresponds to 95% confidence interval.




**Supplementary Figure 5. Metaregression analysis of study relevance score and infarct volume effect size.** Scatterplot shows no correlation between effect size and study relevance score (*P* = 0.2150). The values on the *x*-axis represents study relevance score. Higher score represents higher study relevance. Shaded area corresponds to 95% confidence interval.




**Supplementary Figure 6. Metaregression analysis of study quality score and infarct volume effect size.** Study quality score on the *x*-axis is a sum of risk of bias and study relevance scores. Higher score represents higher study quality. Scatterplot shows no correlation between effect size and study quality score ($P$ = 0.7408). Shaded area corresponds to 95% confidence interval.

**Supplementary table 1.** List of included research articles derived after comprehensive literature search and content evaluation

| # | FULL TITLE | AUTHORS (first 10, et al.) | YEAR | JOURNAL | KEY WORDS (species, virus, gene expression) (Adeno-associated virus-AAV, lentivirus-LV, overexpressed-OE, knockdown-KD) | REFERENCE NUMBER in main text (Not applicable-NA) |
|---|---|---|---|---|---|---|
| 1 | Microglial IRF5-IRF4 regulatory axis regulates neuroinflammation after cerebral ischemia and impacts stroke outcomes* | Al Mamun A, Chauhan A, Qi S, Ngwa C, Xu Y, Sharmeen R, Hazen AL, Li J, Aronowski JA, McCullough LD | 2020 | Proceedings of the National Academy of Sciences of the United States of America | Mouse, LV, OE | [42] |
| 2 | Lentivirally administered glial cell line-derived neurotrophic factor promotes postischemic neurological recovery, brain remodeling and contralesional pyramidal tract plasticity by regulating axonal growth inhibitors and guidance proteins* | Beker M, Caglayan AB, Beker MC, Altunay S, Karacay R, Dalay A, Altintas MO, Kose GT, Hermann DM, Kilic E | 2020 | Experimental Neurology | Mouse, LV, OE | NA |
| 3 | The lncRNA Malat1 functions as a ceRNA to contribute to berberine-mediated inhibition of HMGB1 by sponging miR-181c-5p in poststroke inflammation | Cao D, Liu M, Duan R, Tao Y, Zhou J, Fang W, Zhu J, Niu L, Sun J | 2020 | Acta Pharmacologica Sinica | Mouse, LV, KD+OE | NA |
| 4 | A NeuroD1 AAV-Based Gene Therapy for Functional Brain Repair after Ischemic Injury through In Vivo Astrocyte-to-Neuron Conversion | Chen Y, Ma N, Pei Z, Wu Z,Do-Monte FH, Keefe S, Yellin E, Chen MS, Yin J, Lee G et al. | 2020 | Molecular Therapy | Mouse, AAV, OE | [31] |
| 5 | In vivo Neuroregeneration to Treat Ischemic Stroke Through NeuroD1 AAV-Based Gene Therapy in Adult Non-human Primates | Ge LJ, Yang FH, Li W, Wang T, Lin Y, Feng J, Chen NH, Jiang M, Wang JH,Hu XT et al | 2020 | Frontiers in Cell and Developmental Biology | Primate, AAV, OE | [32] |
| 6 | Intracerebral overexpression of miR-669c is protective in mouse ischemic stroke model by targeting MyD88 and inducing alternative microglial/macrophage activation* | Kolosowska N, Gotkiewicz Maria, Dhungana H, Giudice L, Giugno R, Box D, Huuskonen MT, Korhonen P, Scoyni F, Kanninen KM et al. | 2020 | Journal of Neuroinflammation | Mouse, LV, OE | NA |
| 7 | Restoration of L-OPA1 alleviates acute ischemic stroke injury in rats via inhibiting neuronal apoptosis and preserving mitochondrial function* | Lai Y, Lin P, Chen M, Zhang Y, Chen J, Zheng M, Liu J, Du H, Chen R, Pan X | 2020 | Redox Biology | Rat, AAV, OE | [55] |
| 8 | HDAC2 (Histone deacetylase 2): a critical factor in environmental enrichment-mediated stroke recovery | Lin YH, Yao MC, Wu HY, Dong J, Ni Y, Ko, XL, Chang L, Luo CX, Zhu DY | 2020 | Journal of Neurochemistry | Mouse, AAV and LV, KD | NA |
| 9 | NMMHC IIA Inhibition Ameliorates Cerebral Ischemic/Reperfusion-Induced Neuronal Apoptosis Through Caspase-3/ROCKI/MLC Pathway | Wang G, Wang T, Zhang Y, Li F, Yu B, Kou J | 2020 | Drug Design Development and Therapy | Mouse, AAV, KD | [57] |

| # | FULL TITLE | AUTHORS (first 10, et al.) | YEAR | JOURNAL | KEY WORDS (species, virus, gene expression) (Adeno-associated virus-AAV, lentivirus-LV, overexpressed-OE, knockdown-KD) | REFERENCE NUMBER in main text (Not applicable-NA) |
|---|---|---|---|---|---|---|
| 10 | The neuroprotective and neurorestorative effects of growth differentiation factor 11 in cerebral ischemic injury* | Zhao Y, Wang LH, Peng A, Liu XY, Wang Y, Huang SH, Liu T, Wang XJ, Chen ZY | 2020 | Brain Research | Rat, LV, OE | NA |
| 11 | Ginsenoside Rg1 protects against ischemic/reperfusion-induced neuronal injury through miR-144/Nrf2/ARE pathway* | Chu SF, Zhang Z, Zhou X, He W, Chen C, Luo P, Liu D, Ai Q, Gong HF, Wang Z, Sun H, Feng Z, Chen N | 2019 | Acta Pharmacologica Sinica | Rat, AAV, KD | NA |
| 12 | Lysophosphatidic acid receptor 1 (LPA1) plays critical roles in microglial activation and brain damage after transient focal cerebral ischemia* | Gaire BP, Sapkota A, Song MR, Choi JW | 2019 | Journal of Neuroinflammation | Mouse, LV, OE | NA |
| 13 | Lentiviral-mediated silencing of mast cell-expressed membrane protein 1 promotes angiogenesis of rats with cerebral ischemic stroke | Jian R, Yang M, Xu F | 2019 | Journal of Cellular Biochemistry | Rat, LV, KD | NA |
| 14 | The differential neuroprotection of HSP70-hom gene single nucleotide polymorphisms: In vitro (neuronal hypoxic injury model) and in vivo (rat MCAO model) studies* | Wang Y, Liu Y, Sun K, Wei Y, Fu L, Hou Z, Yi X, Ma D, Wang W, Jin X | 2019 | Gene | Rat, LV, OE | NA |
| 15 | TRIM9-Mediated Resolution of Neuroinflammation Confers Neuroprotection upon Ischemic Stroke in Mice* | Zeng J, Wang Y, Luo Z, Chang LC, Yoo JS, Yan H, Choi Y, Xie X, Deverman BE, Gradinaru V, et al. | 2019 | Cell Reports | Mouse, AAV, OE | [36] |
| 16 | LncRNA-1810034E14Rik reduces microglia activation in experimental ischemic stroke* | Zhang X, Zhu X, Ji B, Cao X, Yu L, Zhang Y, Bao X, Xu Y, Jin J | 2019 | Journal of Neuroinflammation | Mouse, LV, OE | NA |
| 17 | Sema4D/PlexinB1 inhibition ameliorates blood-brain barrier damage and improves outcome after stroke in rats* | Zhou Y, Li Y, Jin H, Wu J, He Q, Wang X, Lei H, Hu B | 2019 | The FASEB Journal | Rat, LV, KD | NA |
| 18 | Expression of nerve growth factor carried by pseudotyped lentivirus improves neuron survival and cognitive functional recovery of post-ischemia in rats | Cao J, Lin Y, Han Y, Ding S, Fan Y, Pan Y, Zhao B, Guo Q, Sun W, Wan J et al. | 2018 | CNS Neuroscience & Therapeutics | Rat, LV, OE | NA |
| 19 | Identification of Sphingosine 1-Phosphate Receptor Subtype 1 (S1P1) as a Pathogenic Factor in Transient Focal Cerebral Ischemia* | Gaire B, Lee C, Sapkota A, Lee SY, Chun J, Cho HJ, Nam T, Choi, Ji W | 2018 | Molecular Neurobiology | Mouse, LV, KD | NA |
| 20 | Ras-Related C3 Botulinum Toxin Substrate 1 Promotes Axonal Regeneration after Stroke in Mice* | Liu L, Yuan H, Yi Y, Koellhoffer EC, Munshi Y, Bu F, Zhang Y, Zhang Z, McCullough LD, Li J | 2018 | Translational Stroke Research | Mouse, LV, OE | NA |

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|---|---|---|---|---|---|---|
| 21 | Astrocyte Autophagy Flux Protects Neurons Against Oxygen-Glucose Deprivation and Ischemic/Reperfusion Injury* | Liu X, Tian F, Wang S, Wang F, Xiong L | 2018 | Rejuvenation Research | Mouse, AAV, OE | [47] |
| 22 | Poststroke delivery of MANF promotes functional recovery in rats | Matlik K, Anttila JE, Kuan-Yin T, Smolander OP, Pakarinen E, Lehtonen L, Abo-Ramadan U, Lindholm P, Zheng C, Harvey B, et al. | 2018 | Science Advances | Rat, AAV, OE | [56] |
| 23 | MicroRNA-365 modulates astrocyte conversion into neuron in adult rat brain after stroke by targeting Pax6 | Mo J, Liu Q, Kou Z, Wu K, Yang P, Chen X, Sun F | 2018 | Glia | Rat, LV, OE | [48] |
| 24 | Survivin overexpression via adeno-associated virus vector Rh10 ameliorates ischemic damage after middle cerebral artery occlusion in rats* | Sehara Y, Inaba T, Urabe T, Kurosaki F, Urabe M, Kaneko N, Shimazaki K, Kawai K, Mizukami H | 2018 | European Journal of Neuroscience | Rat, AAV, OE | NA |
| 25 | Inhibition of proprotein convertase subtilisin/kexin type 9 attenuates neuronal apoptosis following focal cerebral ischemia via apolipoprotein E receptor 2 downregulation in hyperlipidemic mice | Wang L, Wang Z, Shi J, Jiang Q, Wang H, Li X, Hao D | 2018 | International journal of Molecular Medicine | Mouse, LV, KD | [50] |
| 26 | Lentivirus-mediated overexpression of OTULIN ameliorates microglia activation and neuroinflammation by depressing the activation of the NF-κB signaling pathway in cerebral ischemia/reperfusion rats* | Xu H, Qin W, Hu X, Mu S, Zhu J, Lu W, Luo Y | 2018 | Journal of Neuroinflammation | Rat, LV, OE | NA |
| 27 | MicroRNA-195 protection against focal cerebral ischemia by targeting CX3CR1* | Yang G, Liu Z, Wang L, Chen X, Wang X, Dong Q, Zhang D, Yang Z, Zhou Q, Sun J, et al. | 2018 | Journal of Neurosurgery | Mouse, LV, OE | NA |
| 28 | Neuroglobin promotes neurogenesis through Wnt signaling pathway | Yu Z, Cheng C, Liu Y, Liu N, Lo EH, Wang X | 2018 | Cell Death & Disease | Mouse, LV, OE | NA |
| 29 | Netrin-1 Promotes Synaptic Formation and Axonal Regeneration via JNK1/c-Jun Pathway after the Middle Cerebral Artery Occlusion. | Zheng M, Chen R, Chen H, Zhang Y, Chen J, Lin P, Lan Q, Yuan Q, Lai Y, Jiang X, Pan X, Liu N | 2018 | Frontiers in Cellular Neuroscience | Rat, AAV, OE | [25] |
| 30 | Hypoxia Response Element-Regulated MMP-9 Promotes Neurological Recovery via Glial Scar Degradation and Angiogenesis in Delayed Stroke* | Cai H, Ma Y, Jiang L, Mu Z, Jiang Z, Chen X, Wang Y, Yang GY, Zhang Z | 2017 | Molecular Therapy | Mouse, LV, OE | NA |

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|---|---|---|---|---|---|---|
| 31 | G6PD plays a neuroprotective role in brain ischemia through promoting pentose phosphate pathway* | Cao L, Zhang D, Chen J, Qin Y, Sheng R, Feng X, Chen Z, Ding Y, Li M, Qin Z | 2017 | Free Radical Biology & Medicine | Mouse, LV, OE | [64] |
| 32 | Enhanced Proteostasis in Post-ischemic Stroke Mouse Brains by Ubiquilin-1 Promotes Functional Recovery | Liu Y, Qiao F, Wang H | 2017 | Cellular and Molecular Neurobiology | Mouse, LV, OE | NA |
| 33 | Astrocyte-specific insulin-like growth factor-1 gene transfer in aging female rats improves stroke outcomes* | Okoreeh AK, Bake S, Sohrabji F | 2017 | Glia | Rat, AAV, OE | [20] |
| 34 | Raf Kinase Inhibitor Protein Attenuates Ischemic-Induced Microglia Cell Apoptosis and Activation Through NF-κB Pathway* | Su L, Zhang R, Chen Y, Zhu Z, Ma C | 2017 | Cellular Physiology and Biochemistry | Unknown, LV, OE | NA |
| 35 | Matrix Metalloprotease 3 Exacerbates Hemorrhagic Transformation and Worsens Functional Outcomes in Hyperglycemic Stroke* | Hafez S, Abdelsaid M, El-Shafey S, Johnson MH, Fagan, SC, Ergul A | 2016 | Stroke | Rat, LV, KD | NA |
| 36 | Calcium/calmodulin-dependent protein kinase kinase is neuroprotective in stroke in aged mice* | Liu L, Yuan H, Denton K, Li X, McCullough L, Li J | 2016 | European Journal of Neuroscience | Mouse, LV, OE | [37] |
| 37 | HDAC9 exacerbates endothelial injury in cerebral ischaemia/reperfusion injury* | Shi W, Wei X, Wang Z, Han H, Fu Y, Liu J, Zhang Y, Guo J, Dong C, Zhou D, et al. | 2016 | Journal of Cellular and Molecular Medicine | Rat, LV, KD | NA |
| 38 | UHPLC-Q-TOF-MS based serum metabonomics revealed the metabolic perturbations of ischemic stroke and the protective effect of RKIP in rat models* | Su L, Zhao H, Zhang X, Lou Z, Dong X | 2016 | Molecular BioSystems | Rat, LV, OE | NA |
| 39 | MiR-455 inhibits neuronal cell death by targeting TRAF3 in cerebral ischemic stroke* | Yao S, Tang B, Li G, Fan R, Cao F | 2016 | Neuropsychiatric Disease and Treatment | Mouse, LV, KD | NA |
| 40 | Lentivirus-Mediated Overexpression of MicroRNA-210 Improves Long-Term Outcomes after Focal Cerebral Ischemia in Mice | Zeng LL, He XS, Liu JR, Zheng CB, Wang YT, Yang GY | 2016 | CNS Neuroscience and Therapeutics | Mouse, LV, OE | NA |
| 41 | TIGAR contributes to ischemic tolerance induced by cerebral preconditioning through scavenging of reactive oxygen species and inhibition of apoptosis* | Zhou JH, Zhang TT, Song DD, Xia YF, Qin ZH, Sheng R | 2016 | Scientific Reports | Mouse, LV, OE | NA |
| 42 | Hypoxia-controlled matrix metalloproteinase-9 hyperexpression promotes behavioral recovery after ischemia | Cai H, Mu Z, Jiang Z, Wang Y, Yang, GY, Zhang Z | 2015 | Neuroscience Bulletin | Mouse, LV, OE | [40] |

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|---|---|---|---|---|---|---|
| 43 | Role of neuronal NADPH oxidase 1 in the peri-infarct regions after stroke* | Choi DH, Kim JH, Lee KH, Kim HY, Kim YS, Choi WS, Lee J | 2015 | PLoS One | Rat, AAV, KD | [54] |
| 44 | CXCL12 Gene Therapy Ameliorates Ischemia-Induced White Matter Injury in Mouse Brain | Li Y, Tang G, Liu Y, He X, Huang J, Lin X, Zhang Z, Yang G, Wang Y | 2015 | Stem Cells Translational Medicine | Mouse, AAV, OE | NA |
| 45 | Gadd45b Mediates Axonal Plasticity and Subsequent Functional Recovery After Experimental Stroke in Rats | Liu B, Li L, Tan X, Zhang Y, Jiang Y, He G, Chen Q, Li C | 2015 | Molecular neurobiology | Rat, LV, KD | NA |
| 46 | Matrix Metalloproteinase-8 is a Novel Pathogenetic Factor in Focal Cerebral Ischemia* | Han JE, Lee EJ, Moon E, Ryu JH, Choi JW, Kim HS | 2014 | Molecular Neurobiology | Mouse, LV, KD | NA |
| 47 | Neuroprotection by rAAV-mediated gene transfer of bone morphogenic protein 7* | Heinonen A, Rahman M, Dogbevia G, Jakobi H, Wölfl S, Sprengel R, Schwaninger M | 2014 | BMC Neuroscience | Mouse, AAV, OE | NA |
| 48 | The gene silencing transcription factor REST represses miR-132 expression in hippocampal neurons destined to die | Hwang J, Kaneko N, Noh K, Pontarelli F, Zukin RS | 2014 | Journal of Molecular Biology | Rat, LV, OE | NA |
| 49 | A TIGAR-regulated metabolic pathway is critical for protection of brain ischemia* | Li M, Sun M, Cao L, Gu J, Ge J, Chen J, Han R, Qin Y, Zhou Z, Ding Y, Qin Z | 2014 | The Journal of Neuroscience | Mouse, LV, OE | NA |
| 50 | Postacute stromal cell-derived factor-1α expression promotes neurovascular recovery in ischemic mice* | Li Y, Huang J, He X, Tang G, Tang Y, Liu Y, Lin X, Lu Y, Yang G, Wang Y | 2014 | Stroke | Mouse, AAV, OE | [39] |
| 51 | Ectopic expression of human angiopoietin-1 promotes functional recovery and neurogenesis after focal cerebral ischemia* | Meng Z, Li M, He Q, Jiang S, Zhang X, Xiao J, Bai Y | 2014 | Neuroscience | Rat, LV, OE | [26] |
| 52 | Protease activated receptor-1 mediates cytotoxicity during ischemia using in vivo and in vitro models | Rajput PS, Lyden PD, Chen, B, Lamb, JA, Pereira B, Lamb, A, Zhao L, Lei IF, Bai J | 2014 | Neuroscience | Rat, LV, KD | [41] |
| 53 | Netrin-1 overexpression promotes white matter repairing and remodeling after focal cerebral ischemia in mice | He X, Li Y, Lu H, Zhang Z, Wang Y, Yang GY | 2013 | Journal of Cerebral Blood Flow and Metabolism | Mouse, AAV, OE | [24] |
| 54 | Overexpression of adiponectin improves neurobehavioral outcomes after focal cerebral ischemia in aged mice* | Miao J, Shen L, Tang Y, Wang Y, Tao M, Jin K, Zhao Y, Yang G | 2013 | CNS Neuroscience & Therapeutics | Mouse, AAV, OE | [38] |

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|---|---|---|---|---|---|---|
| 55 | Overexpression of adiponectin promotes focal angiogenesis in the mouse brain following middle cerebral artery occlusion | Shen L, Miao J, Yuan F, Zhao Y, Tang Y, Wang Y, Zhao Y, Yang G-Y Y. | 2013 | Gene Therapy | Mouse, AAV, OE | NA |
| 56 | NOX2 deficiency ameliorates cerebral injury through reduction of complexin II-mediated glutamate excitotoxicity in experimental stroke* | Wang Z, Wei X, Liu K, Zhang X, Yang F, Zhang H, He Y, Zhu T, Li F, Shi W, et al. | 2013 | Free Radical Biology and Medicine | Mouse, LV, KD | NA |
| 57 | Akt isoforms differentially protect against stroke-induced neuronal injury by regulating mTOR activities* | Xie R, Cheng M, Li M, Xiong X, Daadi M, Sapolsky RM, Zhao H | 2013 | Journal of Cerebral Blood Flow and Metabolism | Rat, LV, OE | [52] |
| 58 | Local administration of AAV-BDNF to subventricular zone induces functional recovery in stroke rats* | Yu S, Tseng K, Shen H, Harvey BK, Airavaara M, Wang Y | 2013 | PLoS One | Rat, AAV, OE | [60] |
| 59 | MiRNA-424 protects against permanent focal cerebral ischemia injury in mice involving suppressing microglia activation* | Zhao H, Wang J, Gao L, Wang R, Liu X, Gao Z, Tao Z, Xu C, Song J, Ji X, Luo Y | 2013 | Stroke | Mouse, LV, OE | [51] |
| 60 | Netrin-1 hyperexpression in mouse brain promotes angiogenesis and long-term neurological recovery after transient focal ischemia* | Lu H, Wang Y, He X, Yuan F, Lin X, Xie B, Tang G, Huang J, Tang Y, Jin K et al. | 2012 | Stroke | Mouse, AAV, OE | [21] |
| 61 | Induction of autophagy contributes to the neuroprotection of nicotinamide phosphoribosyltransferase in cerebral ischemia* | Wang P, Guan Y, Du H, Zhai Q, Su D, Miao C | 2012 | Autophagy | Rat, LV, OE | NA |
| 62 | siRNA-mediated silence of protease-activated receptor-1 minimizes ischemic injury of cerebral cortex through HSP70 and MAP2* | Zhang J, Wang Y, Zhu P, Wang X, Lv M, Feng H | 2012 | Journal of the Neurological Sciences | Rat, AAV, KD | NA |
| 63 | Role of salt-induced kinase 1 in androgen neuroprotection against cerebral ischemia* | Cheng J, Uchida M, Zhang W, Grafe MR, Herson PS, Hurn PD | 2011 | Journal of Cerebral Blood Flow and Metabolism | Mouse, LV, KD | NA |
| 64 | Targeted over-expression of glutamate transporter 1 (GLT-1) reduces ischemic brain injury in a rat model of stroke* | Harvey BK, Airavaara M, Hinzman J, Wires EM, Chiocco MJ, Howard DB, Shen H, Gerhardt G, Hoffer BJ, Wang Y | 2011 | PLoS One | Rat, AAV, OE | NA |

| # | FULL TITLE | AUTHORS (first 10, et al.) | YEAR | JOURNAL | KEY WORDS (species, virus, gene expression) (Adeno-associated virus-AAV, lentivirus-LV, overexpressed-OE, knockdown-KD) | REFERENCE NUMBER in main text (Not applicable-NA) |
|---|---|---|---|---|---|---|
| 65 | Lentivirus-mediated transfer of MMP-9 shRNA provides neuroprotection following focal ischemic brain injury in rats* | Hu Q, Chen C, Khatibi NH, Li L, Yang L, Wang K, Han J, Duan W, Zhang JH, Zhou C | 2011 | Brain Research | Rat, LV, KD | NA |
| 66 | Overexpression of netrin-1 improves neurological outcomes in mice following transient middle cerebral artery occlusion | Lu H, Wang Y, Yuan F, Liu J, Zeng L, Yang G | 2011 | Frontiers of Medicine | Mouse, AAV, OE | [22] |
| 67 | Coexpression of angiopoietin-1 with VEGF increases the structural integrity of the blood-brain barrier and reduces atrophy volume* | Shen F, Walker EJ, Jiang L, Degos V, Li J, Sun B, Heriyanto F, Young WL, Su H | 2011 | Journal of Cerebral Blood Flow and Metabolism | Mouse, AAV, OE | [28] |
| 68 | AAV-mediated netrin-1 overexpression increases peri-infarct blood vessel density and improves motor function recovery after experimental stroke | Sun H, Le T, Chang TTJJ, Habib A, Wu S, Shen F, Young WL, Su H, Liu J | 2011 | Neurobiology of Disease | Rat, AAV, OE | [23] |
| 69 | Nicotinamide phosphoribosyltransferase protects against ischemic stroke through SIRT1-dependent adenosine monophosphate-activated kinase pathway* | Wang P, Xu T, Guan Y, Tian W, Viollet B, Rui Y, Zhai Q, Su D, Miao C | 2011 | Annals of Neurology | Rat +Mouse, LV, OE | NA |
| 70 | Vascular PPARδ protects against stroke-induced brain injury* | Yin K, Deng Z, Hamblin M, Zhang J, Chen YE | 2011 | Arteriosclerosis, Thrombosis, and Vascular Biology | Mouse, LV, KD | NA |
| 71 | rAAV-mediated delivery of brain-derived neurotrophic factor promotes neurite outgrowth and protects neurodegeneration in focal ischemic model | Zhang J, Yu Z, Yu Z, Yang Z, Zhao H, Liu L, Zhao J | 2011 | International Journal of Clinical and Experimental Pathology | Rat, AAV, OE | *[59]* |
| 72 | Widespread cortical expression of MANF by AAV serotype 7: localization and protection against ischemic brain injury* | Airavaara M, Chiocco MJ, Howard DB, Zuchowski KL, Peranen J, Liu C, Fang S, Hoffer BJ, Wang Y, Harvey, BK et al. | 2010 | Experimental Neurology | Rat, AAV, OE | NA |
| 73 | Angiopoietin 1 counteracts vascular endothelial growth factor-induced blood-brain barrier permeability and alleviates ischemic injury in the early stages of transient focal cerebral ischemia in rats* | Zhao Y, Li Z, Wang R, Wei J, Li G, Zhao H | 2010 | Neurological Research | Rat, AAV, OE | [27] |
| 74 | RNA interference-mediated downregulation of Beclin1 attenuates cerebral ischemic injury in rats* | Zheng, YQ, Liu JX, Li XZ, Xu, Li Xu YG | 2009 | Acta Pharmacologica Sinica | Rat, LV, KD | NA |

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|---|---|---|---|---|---|---|
| 75 | Postischemic IGF-1 gene transfer promotes neurovascular regeneration after experimental stroke* | Zhu W, Fan Y, Hao Q, Shen F, Hashimoto T, Yang G, Gasmi M, Bartus RT, Young WL, Chen Y | 2009 | Journal of Cerebral Blood Flow and Metabolism | Mouse, AAV, OE | [19] |
| 76 | Del-1 gene transfer induces cerebral angiogenesis in mice | Fan Y, Zhu W, Yang M, Zhu Y, Shen F, Hao Q, Young WL, Yang GY, Chen Y | 2008 | Brain Research | Mouse, AAV, OE | NA |
| 77 | Adeno-associated viral vector-mediated hypoxia-regulated VEGF gene transfer promotes angiogenesis following focal cerebral ischemia in mice | Shen F, Fan Y, Su H, Zhu Y, Chen Y, Liu W, Young WL, Yang GY | 2008 | Gene Therapy | Mouse, AAV, OE | [30] |
| 78 | Insulin growth factor-1 gene transfer enhances neurovascular remodeling and improves long-term stroke outcome in mice | Zhu W, Fan Y, Frenzel T, Gasmi M, Bartus RT, Young WL, Yang G, Chen Y | 2008 | Stroke | Mouse, AAV, OE | [18] |
| 79 | Long-lasting regeneration after ischemia in the cerebral cortex* | Leker RR, Soldner F, Velasco I, Gavin DK, Androutsellis-Theotokis A, McKay R | 2007 | Stroke | Rat, AAV, OE | NA |
| 80 | Adeno-associated viral-vector-mediated hypoxia-inducible vascular endothelial growth factor gene expression attenuates ischemic brain injury after focal cerebral ischemia in mice* | Shen F, Su H, Fan Y, Chen Y, Zhu Y, Liu W, Young WL, Yang G | 2006 | Stroke | Mouse, AAV, OE | [29] |
| 81 | Cloning of a novel Apaf-1-interacting protein: a potent suppressor of apoptosis and ischemic neuronal cell death | Cao G, Xiao M, Sun F, Xiao X, Pei W, Li J, Graham SH, Simon RP, Chen J | 2004 | The Journal of Neuroscience | Rat, AAV, OE | [53] |
| 82 | Elevated GDNF levels following viral vector-mediated gene transfer can increase neuronal death after stroke in rats | Arvidsson A, Kirik D, Lundberg C, Madel RJ, Andsberg G, Kokaia Z, Lindvall O | 2003 | Neurobiology of Disease | Rat, AAV, LV, OE | [61] |
| 83 | Anterograde delivery of brain-derived neurotrophic factor to striatum via nigral transduction of recombinant adeno-associated virus increases neuronal death but promotes neurogenic response following stroke | Gustafsson E, Andsberg G, Darsalia V, Mohapel P, Mandel RJ, Kirik D, Lindvall O, Kokaia Z | 2003 | European Journal of Neuroscience | Rat, AAV, OE | [58] |

| # | FULL TITLE | AUTHORS (first 10, et al.) | YEAR | JOURNAL | KEY WORDS (species, virus, gene expression) (Adeno-associated virus-AAV, lentivirus-LV, overexpressed-OE, knockdown-KD) | REFERENCE NUMBER in main text (Not applicable-NA) |
|---|---|---|---|---|---|---|
| 84 | Adeno-associated virus-mediated delivery of BCL-w gene improves outcome after transient focal cerebral ischemia* | Sun Y, Jin K, Clark KR, Peel A, Mao XO, Chang Q, Simon RP, Greenberg DA | 2003 | Gene Therapy | Rat, AAV, OE | [62] |
| 85 | Gene treatment of cerebral stroke by rAAV vector delivering IL-Ira in a rat model* | Tsai TH, Chen SL, Xiao X, Chiang YH, Lin SZ, Kuo SW, Liu DW, Tsao YP | 2003 | NeuroReport | Rat, AAV, OE | [65] |
| 86 | Neuropathological and behavioral consequences of adeno-associated viral vector-mediated continuous intrastriatal neurotrophin delivery in a focal ischemia model in rats | Andsberg G, Kokaia Z, Klein RL, Muzyczka N, Lindvall O, Mandel RJ | 2002 | Neurobiology of Disease | Rat, AAV, OE | [46] |
| 87 | Adeno-associated virus vector-mediated bcl-2 gene transfer into post-ischemic gerbil brain in vivo: prospects for gene therapy of ischemia-induced neuronal death | Shimazaki K, Urabe M, Monahan J, Ozawa K, Kawai N | 2000 | Gene Therapy | Gerbil, AAV, OE | [63] |

*Studies included in meta-analysis.

**Supplementary table 2.** List of candidate genes evaluated as potential stroke treatments

| GENE | KD/OE (Overexpressed-OE, knockdown-KD) | FUNCTION IN STROKE IF KD/OE (Blood brain barrier-BBB) | THERAPEUTIC EFFECT AND OTHER EFFECT (Infarct volume-IV, neurobehavioral deficits-ND, caspase 3-cas3, caspase 9-cas9, Blood-brain barrier-BBB, microvascular density-MVD, neural progenitor cells-NPCs, subventricular zone-SVZ, oligodendrocyte progenitor cells-OPCs, hypoxia response element-HRE, extracellular matrix-ECM, nuclear factor kappa B-NFkB, cerebral blood flow-CBF) | REFERENCE |
|---|---|---|---|---|
| *Aip* | OE | Antiapoptotic | Decreases neuronal apoptosis (decreases activation of cas-3 and cas-9) | Cao 2004 |
| *Akt* | OE | Antiapoptotic | Reduces IV and neuronal apoptosis | Xie 2013 |
| *Angpt1* | OE | Angiogenesis, neurogenesis | Reduces IV, reduces ND, protects BBB integrity, increases MVD | Zhao 2010, Shen 2011, Meng 2014 |
| *Apn* | OE | Angiogenesis, anti-inflammatory | Reduces IV, reduces ND and promotes angiogenesis | Shen 2012, Miao 2013 |
| *Atg7* | OE | Anti-inflammatory | Reduces IV, no changes in ND, decreases neuronal apoptosis | Liu 2018 |
| *Bcl2* | OE | Antiapoptotic | Decreases neuronal apoptosis | Shimazaki 2000 |
| *Bcl2l2* | OE | Antiapoptotic | Reduces IV, reduces ND | Sun 2003 |
| *Bdnf* | OE | Neuroregenerative, neuroprotective | Reduces IV, reduces ND, increases neuronal survival, increases migration of NPCs in SVZ | Andsberg 2002, Gustafsson 2003, Zhang 2011, Yu 2013 |
| *Becn1* | KD | Inhibition of autophagy | Reduces IV, reduces ND, reduces neuronal apoptosis | Zheng 2009 |
| *Bmp7* | OE | Neurogenesis, neuroregenerative | Reduces IV, reduces cell apoptosis | Heinonen 2014 |
| *Ngf* | OE | Neurogenesis | Reduces ND, decreases neuronal apoptosis, promotes axonogenesis and synaptic remodeling | Andsberg 2002, Cao 2018 |
| *Camkkβ* | OE | Anti-inflammatory, BBB protection | Reduces IV, reduces ND | Liu 2016 |
| *Cplx2* | KD | Reduces glutamate-mediated excitotoxicity | Reduces IV, reduces ND and ischemia induced glutamate release processes | Wang 2013 |
| *Cxcl12* | OE | Neurogenesis, angiogenesis, remyelination | Reduces ND, promotes NPCs and OPCs migration and maturation | Li 2014 |
| *Del1* | OE | Angiogenesis | Increases vascular density and promotes endothelial cell proliferation | Fan 2008 |
| *Fgf2* | OE | Angiogenesis and neurogenesis | No changes in IV, reduces ND, increases number of NPCs and immature neurons | Leker 2007 |
| *G6pd* | OE | Antioxidative | Reduces IV and edema, reduces ND and oxidative stress | Cao 2017 |
| *Gadd45b* | KD | Inhibition of neuroregeneration | Increases ND and inhibits axonal regeneration and reorganization | Liu 2015, Beker 2020 |
| *Gdnf* | OE | Neurogenesis, antiapoptotic | No changes in ND and neuronal survival | Arvidsson 2003 |

| GENE | KD/OE (Overexpressed-OE, knockdown-KD) | FUNCTION IN STROKE IF KD/OE (Blood brain barrier-BBB) | THERAPEUTIC EFFECT AND OTHER EFFECT (Infarct volume-IV, neurobehavioral deficits-ND, caspase 3-cas3, caspase 9-cas9, Blood-brain barrier-BBB, microvascular density-MVD, neural progenitor cells-NPCs, subventricular zone-SVZ, oligodendrocyte progenitor cells-OPCs, hypoxia response element-HRE, extracellular matrix-ECM, nuclear factor kappa B-NFkB, cerebral blood flow-CBF) | REFERENCE |
|---|---|---|---|---|
| *Gdf11* | OE | Neuroprotective and neurorestorative | Reduces IV, reduces ND | Zhao 2020 |
| *Glt-1* | OE | Neuroprotection | Reduces IV, reduces ND and decreases glutamate overflow, reduces cell death | Harvey 2011 |
| *Hdac2* | KD | Enhance neuroplasticity | Promotes functional recovery from stroke | Lin 2020 |
| *Hdac9* | KD | Anti-inflammatory, BBB protection | Reduces IV and edema, reduces ND, reduces BBB permeability | Shi 2016 |
| *Hsp70* | OE | Antiapoptotic | Reduces IV, decreases expression of Bax protein | Wang 2019 |
| *Igf1* | OE | Neurogenesis, angiogenesis, anti-inflammatory | Reduces IV, reduces ND, increases MVD, enhances neurogenesis, increases number of NPCs | Zhu 2008, Zhu 2009, Okoreeh 2017 |
| *Il1rn* | OE | Anti-inflammatory | Reduces IV, reduces apoptosis | Tsai 2003 |
| *Irf4* | OE | Anti-inflammatory | Reduces IV, reduces ND | Mamun 2020 |
| *Irf5* | OE | Proinflammatory | Increases IV, increases ND, enhances M1 phenotype, promotes proinflammatory response | Mamun 2020 |
| lncRNA-1810034E14Rik | OE | Anti-inflammatory | Reduces IV, ND, reduces inflammatory response and microglial activation | Zhang 2019 |
| *Malat1* | OE | OE-Proinflammatory | Promotes inflammatory cytokine release | Cao 2020 |
| *Malat1* | KD | KD-Anti-inflammatory | Promotes anti-inflammatory cytokine release | Cao 2020 |
| M*anf* | OE | Neuroprotection | No changes of IV, reduced ND, increases the number of phagocytic cells | Matlik 2018 |
| *Mcemp1* | KD | Angiogenesis, antiapoptotic | Decreases neuronal apoptosis and promotes angiogenesis | Jian 2019 |
| miR-132 | OE | Neurogenesis | Reduces neuronal cell death | Hwang 2014 |
| miR-144 | KD | Increases oxidative stress | Reduces expression of downstream antioxidative genes | Chu 2019 |
| miR-195 | OE | Anti-inflammatory | Reduces IV, reduces ND, downregulates proinflammatory cytokines | Yang 2018 |
| miR-210 | OE | Neurogenesis, angiogenesis | Reduces IV, reduces ND, increases MVD and number of NPCs | Zeng 2016 |
| miR-424 | OE | Anti-inflammatory, antiapoptotic | Reduces IV, reduces apoptosis and microglial activation | Zhao 2013 |
| miR-669c | OE | Anti-inflammatory | Reduces IV, reduces ND | Kolosowska 2020 |
| *Mmp-3* | KD | BBB protection | No change in IV and edema, reduces ND | Hafez 2016 |
| *Mmp-8* | KD | Anti-inflammatory, BBB protection | Reduces IV, reduces microglial activation and cell death | Han 2014 |

| GENE | KD/OE (Overexpressed-OE, knockdown-KD) | FUNCTION IN STROKE IF KD/OE (Blood brain barrier-BBB) | THERAPEUTIC EFFECT AND OTHER EFFECT (Infarct volume-IV, neurobehavioral deficits-ND, caspase 3-cas3, caspase 9-cas9, Blood-brain barrier-BBB, microvascular density-MVD, neural progenitor cells-NPCs, subventricular zone-SVZ, oligodendrocyte progenitor cells-OPCs, hypoxia response element-HRE, extracellular matrix-ECM, nuclear factor kappa B-NFkB, cerebral blood flow-CBF) | REFERENCE |
|---|---|---|---|---|
| *Mmp-9* | KD (acute) | BBB protection | Reduces IV, reduces ND, reduces BBB breakdown | Hu 2011, Yin 2011 |
| *Mmp-9* | OE (sub-acute) | Glial-scar degradation, angiogenesis | Reduces IV, reduces ND (sub-acute phase, HRE controlled), increases ECM remodeling and MVD | Cai 2015, Cai 2017 |
| *Myh9* | KD | Antiapoptotic | Reduces ND, reduced cell death | Wang 2020 |
| *Nampt* | OE | Promote autophagy, anti-energetic depletion | Reduces IV, reduces ND, promotes beneficial autophagy and anti-apoptotic gene expression | Wang 2011, Wang 2012 |
| *Ntn1* | OE | Antiapoptotic, angiogenesis, neurogenesis, axonal regeneration, restores myelination | Reduces IV, reduces ND, increases MVD, OPCs proliferation and synapsis formation | Sun 2011, Lu 2011, Lu 2012 |
| *Neurod1* | OE | Neurogenesis, astrocyte to neuron transition | Promotes transition from active astrocytes to mature neurons | Chen 2020, Ge 2020 |
| *Ngb* | OE | Neurogenesis | Promotes cellular proliferation and neurogenesis | Yu 2018 |
| *Ngf* | OE | Neurogenesis | Reduces ND, increases neural survival | Andsberg 2002 |
| *Nox1* | KD | Antioxidative, antiapoptotic | Reduces IV, reduces ND, reduced cell apoptosis and superoxide generation | Choi 2015 |
| *Opa1* | OE | Antioxidative, antiapoptotic | Reduces IV, reduces ND, preserves mitochondrial structure, and reduces neuronal apoptosis | Lai 2020 |
| *Otulin* | OE | Anti-inflammatory | Reduces IV, reduces ND, prevents microglia accumulation and activation | Hu 2018 |
| *Par1* | KD | Neuroprotective | Reduces IV, reduces ND | Zhang 2012, Rajput 2014 |
| *Pax6* | OE | Neurogenesis, astrocyte to neuron transition | Promotes reactive astrocyte to neuron conversion | Mo 2018 |
| *Pcsk9* | KD | Antiapoptotic | Reduces IV and neuronal cell death | Wang 2018 |
| *PlexinB1* | KD | Anti-inflammatory, BBB protection | Reduces IV, reduces ND, reduces vascular leakage and BBB breakdown | Zhou 2018 |
| *Rac1* | OE | Neuroregeneration, axonal outgrowth | Reduces ND, increases axonal density and reduces glial-scar formation | Liu 2018 |
| *Rkip* | OE | Anti-inflammatory, antiapoptotic | Reduces IV, reduces cell death and microglia activation | Su 2017, Su 2016 |
| *S1p1* | KD | Anti-inflammatory, BBB protection | Reduces IV, reduces ND, reduces cell death, microglia activation and BBB breakdown | Gaire 2018 |
| *Sik 1* | KD | Neurodetrimental | Increases IV, exacerbates neuronal cell death | Cheng 2011 |

| GENE | KD/OE (Overexpressed-OE, knockdown-KD) | FUNCTION IN STROKE IF KD/OE (Blood brain barrier-BBB) | THERAPEUTIC EFFECT AND OTHER EFFECT (Infarct volume-IV, neurobehavioral deficits-ND, caspase 3-cas3, caspase 9-cas9, Blood-brain barrier-BBB, microvascular density-MVD, neural progenitor cells-NPCs, subventricular zone-SVZ, oligodendrocyte progenitor cells-OPCs, hypoxia response element-HRE, extracellular matrix-ECM, nuclear factor kappa B-NFkB, cerebral blood flow-CBF) | REFERENCE |
|---|---|---|---|---|
| *Survivin* | OE | Antiapoptotic | Reduces IV, reduces cell apoptosis | Sehara 2018 |
| *Tigar* | OE | Antioxidative, antiapoptotic | Reduces IV and edema, reduces ND, reduces oxidative stress and apoptosis | Zhou 2016, Li 2014 |
| *Traf3* | KD | Neuroprotective | Reduces IV, reduces cell death | Yao 2016 |
| *Trim9* | OE | Anti-inflammatory | Reduces IV, reduces cell death and neuroinflammation by NFkB pathway inhibition | Zeng 2019 |
| *Ubq1n* | OE | Neuroprotective | Reduces ND, reduces accumulation of protein aggregates | Liu 2018 |
| *Vegfa* | OE | Angiogenesis, antiapoptotic, neurogenesis | Reduces IV, ND, reduces cell apoptosis, increases MVD, recovers CBF | Zhao 2010, Shen 2006, Shen 2008, Shen 2011 |